

1. $32,767 = 31 * 1,057$

(a) 2. It is both a number greater than one and is able to factor into the given number.

2. (a) There are no values of 'n' for which $3^n - 1$ is a prime.

x	Is it prime?	$3^x - 1$	Is it Prime?
1	yes	\$2	no (even)
2	yes	\$8	no (even)
3	yes	\$26	no (even)
4	no	\$80	no (even)
5	yes	\$242	no (even)
6	no	\$728	no (even)
7	yes	\$2186	no (even)
8	no	\$6560	no (even)
9	yes	\$19682	no (even)
10	no	\$59048	no (even)

(b) When n is prime, then $3^n - 2^n$ might be prime.

Table 1: An example table using booktabs.

x	Is it prime?	$3^x - 2^n$	Is it Prime?
1	yes	\$1	yes
2	yes	\$5	yes
3	yes	\$19	yes
4	no	\$65	no (divisible by 5)
5	yes	\$211	yes
6	no	\$665	no (divisible by 5)
7	yes	\$2059	no (divisible by 29)
8	no	\$6305	no (divisible by 5)
9	yes	\$19171	no (divisible by 19)
10	no	\$58025	no (divisible by 5)

3. (a) $(2 * 3 * 5 * 7) + 1 = 211$ This is a prime number because there are no factors below $\sqrt{211} < 15$ except 1.

(b) $(2 * 5 * 11) + 1 = 111 // 111 / 3 = 37$ 37 is a prime number because there are no factors below $\sqrt{37} < 7$ except 1.

4. $(5 + 1)! + 2 = 720$ The requested sequence is: 722, 723, 724, 725, and 726
 $722/2 = 361$

$$723/3 = 241$$

$$724/4 = 181$$

$$725/5 = 145$$

$$726/6 = 121$$

5. (a) $2^6(2^7 - 1) = 8128$

(b) $2^{10}(2^{11} - 1) = 2096128$

6. No, because that would require one of the numbers to be a multiple of three. If the first number 'x' would be such that $x/3 = y, r1$, then the next number must be a multiple of three, $(1 + 2 = 3)$ gets rid of the remainder. If the first number is such that the remainder would be two, the next one would have a remainder of 1 $(2 + 2 = 4, 4/3 = 1, r1)$, the third one would then be a multiple of 3.

7. Of 220: $1 + 2 + 4 + 5 + 10 + 11 + 20 + 22 + 44 + 55 + 110 = 228$

Of 228: $1 + 2 + 4 + 71 + 142 = 220$